

Structural Social Justice Metric (SSJ): Operationalizing the Distribution of Chaos

Negentropic Justice posits that true equity is measured by the equitable access to order (Negentropy) and the just distribution of inevitable disorder (Entropy). The objective of this Metric is to map the vulnerability of an individual or collective (the Demand Vector) against the stability provided by the system (the Supply Vector).

I. Defining the Entropic Dimensions (The Vectors)

The Negentropic Justice Metric (NJM) operates in an N-dimensional vector space. Each dimension (axis) represents a critical source of chaos or systemic instability whose distribution must be equitable.

We propose six initial structural dimensions (Axes $\mathbf{X}_1$ to $\mathbf{X}_6$):

AXIS	DIMENSION	DESCRIPTION OF CHAOS (ENTROPY)	KEY MEASURE OF ORDER (NEGENTROPY)
$\mathbf{X}_1$	Infrastructure	Frequency of failures in essential services (water, electricity, transportation).	Network stability rate (Uptime) and Redundancy (number of alternative routes).
$\mathbf{X}_2$	Vulnerability	Exposure to environmental and climatic risk (floods, pollution, toxicity).	Per capita investment in climate defenses and environmental cleanliness index.
$\mathbf{X}_3$	Social Capacity	Frequency of collapse in health, education, or public safety systems.	Average response time and access to high-efficiency education.
$\mathbf{X}_4$	Disruption/Tech	Exposure to rapid job obsolescence and access to technological capital.	Rate of sponsored labor reconversion and access to digital infrastructure.
$\mathbf{X}_5$	Legal Integrity	Frequency of judicial failures due to corruption or inefficiency.	Index of trust in the legal system and dispute resolution time.
$\mathbf{X}_6$	Fiscal Sustainability	Exposure to hyperinflation, unsustainable public debt, or economic crises.	Stability of purchasing power and control of short-term debt.

II. The Structural Mapping: Demand Vector vs. Supply Vector

To determine "justice," two vectors are compared in this N-dimensional space.

1. Supply Vector ($\mathbf{V}_S$) - The Order Produced by the System

This vector represents the Negentropy or stability that the State or the market supplies to society. It is composed of the direct order metrics for each dimension (e.g., S_1 would be the power grid Uptime, normalized from 0 to 1).

$$\mathbf{V}_S = [S_1, S_2, S_3, S_4, S_5, S_6]$$

2. Demand Vector ($\mathbf{V}_D$) - The Chaos Assumed by the Citizen

This vector represents the vulnerability or chaos that an individual or collective must assume. It is measured through surveys that evaluate the perceived risk and the capacity to absorb shocks in each dimension.

Measurement: Perceived risk and resilience capacity (inverse to vulnerability) are normalized from 0 to 1.

$$\mathbf{V}_D = [D_1, D_2, D_3, D_4, D_5, D_6]$$

III. The Central Calculation: Entropic Distance (Cosine Similarity)

Justice is measured by the distance or, more precisely, the Angle between these two vectors. Cosine Similarity (θ) measures the directional alignment of the vectors.

$$\text{Similarity}(\mathbf{V}_S, \mathbf{V}_D) = \frac{\mathbf{V}_S \cdot \mathbf{V}_D}{\|\mathbf{V}_S\| \cdot \|\mathbf{V}_D\|} = \cos(\theta)$$

Structural Interpretation:

- $\cos(\theta) \approx 1$ (Small Angle): **High Negentropic Justice.** The order the system produces ($\mathbf{V}_S$) is exactly the order the collective needs and demands ($\mathbf{V}_D$).
- $\cos(\theta) \approx 0$ (90° Angle): **Extreme Misalignment.** The system ignores the most critical sources of chaos for the citizen.
- $\cos(\theta) \approx -1$ (Large Angle): **Active Entropic Injustice.** The system's order is actively creating more chaos or vulnerability.

IV. Scalable Application: The Injustice of Externalization

Injustice occurs when the actor benefiting from the generated entropy ($\mathbf{V}_S$ low) does not bear the cost of the Negentropy required to restore order ($\mathbf{V}_D$ high).

1. The Party Chaos (Social Injustice)

CONCEPT	ENTROPIC LOAD ANALYSIS	DEMAND VECTOR ($\mathbf{V}_D$)	RESULT (IPE)
Chaos Generator	The group of 10 guests.	N/A (Benefit).	$\cos(\theta) \approx 0$
Vulnerable Actor	The 2 people who clean.	High in $\mathbf{X}_3$ (Social Capacity/Energy). Their demand for Negentropy (rest, order) is ignored.	Low IPE. The social system ignores the energetic load placed upon the vulnerable actor.

2. Domestic Chaos (Structural Injustice)

CONCEPT	ENTROPIC LOAD ANALYSIS	DEMAND VECTOR ($\mathbf{V}_D$)	RESULT (IPE)
Chaos Generator	Children.	N/A (Play, Benefit).	$\cos(\theta) \approx 0$
Vulnerable Actor	The father/mother.	High in $\mathbf{X}_1$ (Infrastructure/Recovery). The chaos drains their post-work resilience capacity.	Low IPE. The family system does not supply the necessary Negentropy to protect the infrastructure (health and rest) of its primary operator.

3. Political Campaign Chaos (Civic Injustice)

CONCEPT	ENTROPIC LOAD ANALYSIS	DEMAND VECTOR ()	RESULT (IPE)
Chaos Generator	Political Parties.	N/A (Visibility, Electoral Benefit).	$\cos(\theta) \approx 0$

CONCEPT	ENTROPIC LOAD ANALYSIS	DEMAND VECTOR ($\mathbf{V}_D$)	RESULT (IPE)
Vulnerable Actor	Citizenship/City Hall.	High in $\mathbf{X}_2$ (Environmental Vulnerability) and $\mathbf{X}_6$ (Fiscal Cost). City Hall resources (Financial Negentropy) are diverted from other needs.	Low IPE. The cost of political Entropy is externalized. The party benefits without fulfilling the mandatory Negentropy (cleaning).

4. The Entropic Cost of the Demonstration (The Final Disposal Problem)

CONCEPT	ENTROPIC LOAD ANALYSIS	DEMAND VECTOR ($\mathbf{V}_D$)	RESULT (IPE)
Chaos Generator	Protesters (exercising a legitimate right).	N/A (Freedom of expression).	$\cos(\theta) \approx 0.1$ to 0.5
Vulnerable Actor	City Hall/Non-Participating Citizens.	High in $\mathbf{X}_6$ (Fiscal Cost). City Hall must bear the cost of the "final disposal" of the disorder.	Low/Medium IPE. The low IPE indicates that, although the chaos is legitimate, the lack of a Negentropy plan (cleanup) by the organizers imposes an unfair burden on the system.

Conclusion on Entropic Load

Your observation is the cornerstone of Negentropic Justice:

Ultimately, in human actions, people worry about the reactions (linear consequence) but never about the final disposal of the generated disorder (waste, consequence of higher order).

This is the exact definition of the **Injustice of Externalization**. Modern justice does not fail in the redistribution of money, but in the redistribution of the entropic reversal load (the work and energy to clean up the chaos).

The NJM forces us to calculate the cost of the "final disposal of disorder" and ensure that the Supply Vector ($\mathbf{V}_S$) of the causal actor is aligned with the Demand Vector ($\mathbf{V}_D$) of the vulnerable actor.

Mathematical Appendix: Cosine Similarity as a Metric for Structural Justice

I. Introduction to Vector Alignment

The Negentropic Justice Metric (NJM) uses the concept of Cosine Similarity to determine the coherence between the order that the system supplies ($\mathbf{V}_S$) and the chaos that the citizen demands to assume ($\mathbf{V}_D$). Instead of measuring the absolute magnitude of order (how much there is), we measure the direction (whether what is supplied is what is truly needed).

The scalar product (or dot product) is the foundation of this measurement.

II. The Scalar Product ($\mathbf{V}_S \cdot \mathbf{V}_D$)

The scalar product is an operation that takes two vectors and returns a scalar. Geometrically, it measures the projection of one vector onto the other. It is defined in two ways:

A. Algebraic Definition (in 2D)

Let us consider a two-dimensional space where the **Supply Vector** ($\mathbf{V}_S$) represents (For example: X_1 = Infrastructure Stability, X_2 = Environmental Cleanliness) and the **Demand Vector** ($\mathbf{V}_D$) represents the perceived vulnerability.

Clarification on Components: The values S_i and D_i are assumed to be **normalized indices** (dimensionless). This is crucial to ensure that different dimensions (X_1, X_2) are comparable, regardless of their original units of measurement (e.g., money, time, or physical quantity).

$$\mathbf{V}_S = (S_1, S_2)$$

$$\mathbf{V}_D = (D_1, D_2)$$

The scalar product is calculated as the sum of the products of their components:

$$\mathbf{V}_S \cdot \mathbf{V}_D = S_1D_1 + S_2D_2$$

B. Geometric Definition

The scalar product is also related to the magnitude of the vectors ($\|\mathbf{V}\|$) and the cosine of the angle (θ) between them:

$$\mathbf{V}_S \cdot \mathbf{V}_D = \|\mathbf{V}_S\| \cdot \|\mathbf{V}_D\| \cdot \cos(\theta)$$

III. Cosine Similarity ($\cos(\theta)$)

By solving for $\cos(\theta)$ from the geometric formula, we obtain the **Cosine Similarity**, which measures the directional alignment, independent of the vectors' length:

$$\cos(\theta) = \frac{\mathbf{V}_S \cdot \mathbf{V}_D}{\|\mathbf{V}_S\| \cdot \|\mathbf{V}_D\|}$$

Interpretation for Negentropic Justice

The value of $\cos(\theta)$ ranges between -1 and 1 . This metric allows us to evaluate the fairness of chaos distribution:

- $\cos(\theta) = 1$: **Perfect Alignment (Maximum Justice)**. The angle $\theta = 0^\circ$. The supplied order vector ($\mathbf{V}_S$) points exactly in the same direction as the vulnerability or demand for order vector ($\mathbf{V}_D$). The system is precisely solving the chaos that most affects the citizen.
- $\cos(\theta) = 0$: **Orthogonality (Total Misalignment)**. The angle $\theta = 90^\circ$. The vectors are perpendicular. The system is generating order in totally irrelevant dimensions, leaving critical vulnerabilities unaddressed.
- $\cos(\theta) = -1$: **Opposition (Active Injustice)**. The angle $\theta = 180^\circ$. The system is producing order in a direction that is actually exacerbating the citizen's chaos or vulnerability.

Example of Perfect Alignment (Justice $\cos(\theta) = 1$)

Let's assume a community (the Vulnerable Actor) has a high demand for order in environmental cleanliness ($D_1 = 5$) and a low demand in infrastructure stability ($D_2 = 1$), resulting in $\mathbf{V}_D = (5, 1)$.

If the municipality (the System) focuses its efforts on cleanliness (Negentropy Supply $S_1 = 10$) and little on infrastructure ($S_2 = 2$), then $\mathbf{V}_S = (10, 2)$.

- Scalar Product: $\mathbf{V}_S \cdot \mathbf{V}_D = (5)(10) + (1)(2) = 52$
- Magnitude of $\mathbf{V}_S$: $\|\mathbf{V}_S\| = \sqrt{10^2 + 2^2} = \sqrt{104}$
- Magnitude of $\mathbf{V}_D$: $\|\mathbf{V}_D\| = \sqrt{5^2 + 1^2} = \sqrt{26}$
- Cosine Similarity: $\cos(\theta) = \frac{52}{\sqrt{104} \cdot \sqrt{26}} = \frac{52}{52} = 1$

The result $\cos(\theta) = 1$ confirms that the order supplied by the municipality is **perfectly aligned** with the community's demand for order.